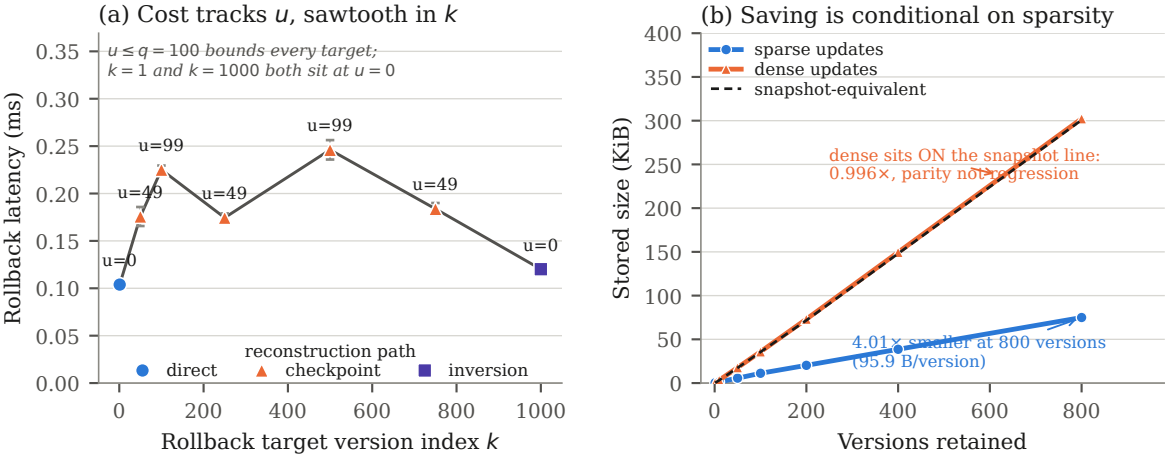
(a) Cost tracks u, sawtooth in k
u ≤ q = 100 bounds every target;
k = 1 and k = 1000 both sit at u = 0
Rollback latency (ms)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
u=0
u=49
u=99
u=49
u=99
u=49
u=0
reconstruction path
direct
checkpoint
inversion
0
200
400
600
800
1000
Rollback target version index k
(b) Saving is conditional on sparsity
Stored size (KiB)
400
350
300
250
200
150
100
50
0
sparse updates
dense updates
snapshot-equivalent
dense sits ON the snapshot line:
0.996×, parity not regression
4.01× smaller at 800 versions
(95.9 B/version)
0
200
400
600
800
Versions retained